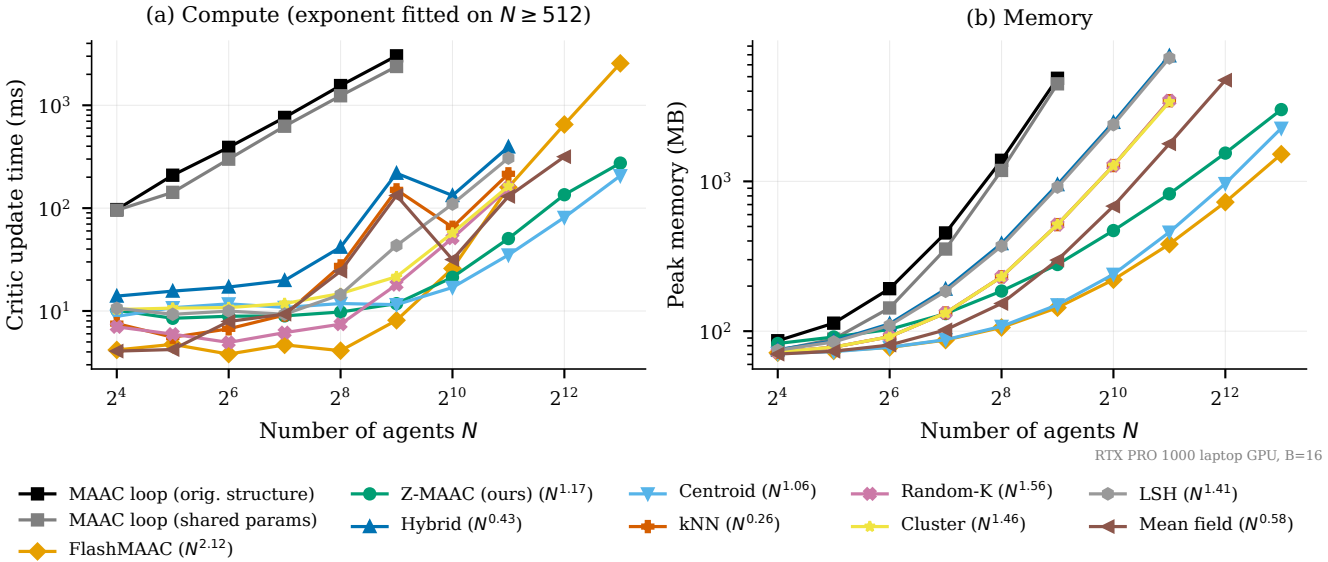
(a) Compute (exponent fitted on $N \geq 512$)
Critic update time (ms)
10^3
10^2
10^1
2^4
2^6
2^8
2^{10}
2^{12}
Number of agents N
(b) Memory
Peak memory (MB)
10^3
10^2
2^4
2^6
2^8
2^{10}
2^{12}
Number of agents N
RTX PRO 1000 laptop GPU, B=16
MAAC loop (orig. structure)
MAAC loop (shared params)
FlashMAAC ($N^{2.12}$)
Z-MAAC (ours) ($N^{1.17}$)
Hybrid ($N^{0.43}$)
Centroid ($N^{1.06}$)
kNN ($N^{0.26}$)
Random-K ($N^{1.56}$)
Cluster ($N^{1.46}$)
LSH ($N^{1.41}$)
Mean field ($N^{0.58}$)